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In this final project, we constructed a prototype of a mechanism that has at least two inputs and at least two outputs. Our mechanism's purpose is to automatically store and collect rainwater when rain is detected by opening the lid to the container. When rain, or moisture, is detected, the hatch is opened to collect rainwater. We also have an LED, serving as our second output, which turns on at the same time. This is to show the user the mechanism is working, even if the container is not in view. When rain is not detected, the hatch remains closed, and the LED is turned off. We also added an override switch to manually open and close the hatch even without water being detected. The motor we used for this mechanism was the stepper motor. We constructed the container out of cardboard and popsicle sticks, gluing them together with hot glue. The lid is made out of a light-weight cardboard to allow for smooth movement. We also added supports made of popsicle sticks and tape to ensure fluidity.

The first input we used was our homemade custom IED sensor mask, which served as our main sensor for sensing rain. The second input we used was the rain sensor, which served as an alternate sensor for sensing rain. The third input was a manual switch, which we could use to override the sensors with the push of a button. The two outputs were the movement of the stepper motor and the LED turning on. They both activated when any of the three inputs were detected.

The first day we worked on the project, we assembled materials and began programming and wiring. At first, we tried using the servo to move the lid to our container, but it didn't have enough power to move it. So, we pivoted a motor, and then to the stepper, which worked much better. Working together, we started to build the prototype water collection container and lid and started planning out the presentation. For the code, we made three iterations of our program. In the first iteration of the program, we made sure that we could move the servo 90 degrees with the manual switch. In the second iteration of the program, we changed the servo to a motor, fixing the wiring. In the third iteration of the program, we changed the motor to a stepper and wired accordingly. At the end of the day, we made sure the LED turned on when the switch was pressed. We also finished building the base of our collection container out of popsicle sticks and cardboard, ensuring the lid can be moved with the stepper.

On the second day of the project, we wired the water sensor and updated the code, making sure it moved the lid when 1.5 volts or more were detected and turned on the LED. We also added onto the prototype water collection container and lid to support the stepper and lid by constructing scaffolding. We continued working on the presentation, adding more information and polishing what was already added. For the

programs, we made iterations four, five, and six. In the fourth iteration of the program, we connected the water sensor to the breadboard, making sure it works as intended. In iterations five and six, we continued adjusting the resistance measurements to ensure the stepper only moved when water was detected. We also made sure the LED turned on when water was sensed. We ended the day with testing our prototype, including programming a resistance test and a breath test. We also wrote a program that recalibrates our stepper when the angle of movement is inaccurate.

On the last day we had to work on the project, we finished the supports for the lid, ensuring smooth movement. We also checked all of the final code, making sure the programs worked accurately and reliably. Finally, we finished the presentation, adding a conclusion and pictures of the prototype, and began cleaning up the station. We didn't make any more iterations of the program, instead just updating our AI use to the programs lacking it. The testbreath.py program was utilized in our testing, as well as the recalibrate.py program. We finished the prototype with time to spare to continue testing and making any tweaks as needed. Lastly, we finished cleaning up our station and began preparing for our demo the next class.

Overall, this project was fun and very educational. I learned a lot about circuits and how to wire using an Arduino, something I knew absolutely nothing about before stepping into the lab. I appreciated all the support from my group members, as they had some experience and helped me understand what we were supposed to do and how we were supposed to do it. I feel like my group worked well together, getting everything done efficiently and masterfully.